EXP.NO: 4	Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.
DATE:26/02/2025	

AIM:

To create a web-based Online Quiz Application that accepts user details (name and quiz answers) through an HTML form and uses a Servlet to process the data and display the quiz result (score) back on the browser.

ALGORITHM:

1. Input:

- The user opens the quiz form in a browser
- Enters their name and selects answers for quiz questions
- Submits the form

2. Form Handling and Servlet Processing:

1. Create the Web Form
- Design an HTML form with:
  - Name input field
  - Multiple-choice quiz questions
- Add a Submit button
- Set form action to Servlet URL and method to POST

2. Create the Servlet Class
- Create a class extending HttpServlet
- Override the doPost() method
- Use request.getParameter() to retrieve:
  - User name
  - Selected answers

3. Process the Data in Servlet
- Compare user answers with correct answers
- Calculate total score
- Store result

4. Send Response to Browser
- Set response type to text/html
- Display:
  - User name
  - Score (e.g., “You scored 2/3”)

5. Deploy the Servlet
- Compile the servlet
- Deploy in Apache Tomcat
- Map using annotation (@WebServlet)

6. Test the Application
- Open browser
- Enter details and attempt quiz
- View result page

Program:

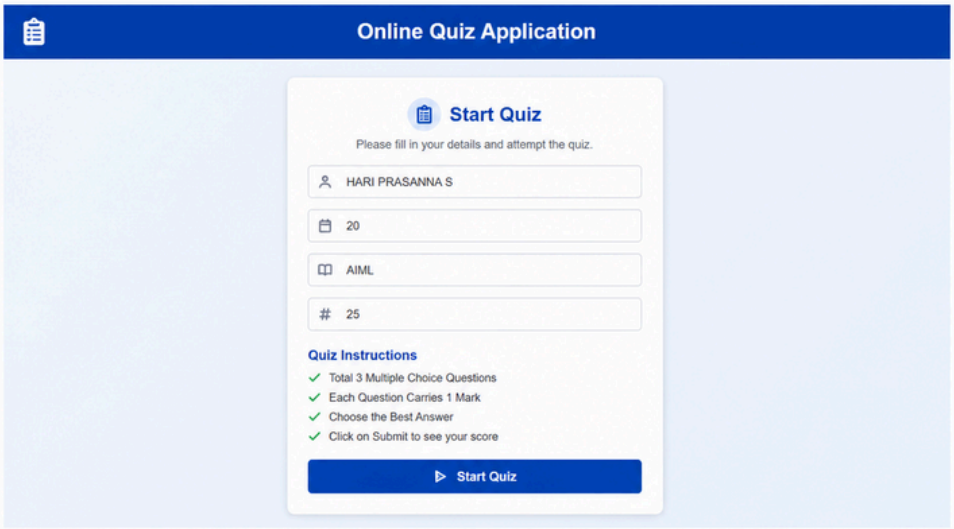
QuizForm.html

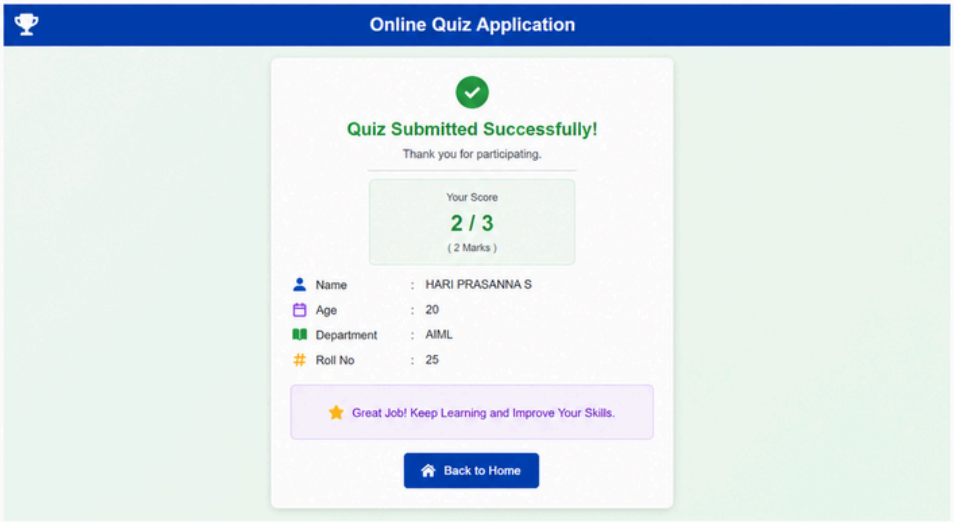
```
<!DOCTYPE html>
<html>
<head>
 <title>Online Quiz</title>
 <style>
 body {
 font-family: Arial;
 background-color: #f2f6ff;
 }
 .container {
 width: 400px;
 margin: 80px auto;
 background: white;
 padding: 20px;
 border-radius: 10px;
 box-shadow: 0 0 10px #ccc;
 }
 h2 {
 text-align: center;
 color: #004aad;
 }
 button {
 width: 100%;
 padding: 10px;
 background: #004aad;
 color: white;
 border: none;
 }
 </style>
</head>
<body>

<div class="container">
 <h2>Online Quiz</h2>

 <form action="QuizServlet" method="post">

 <input type="text" name="name" placeholder="Enter Your Name" required>

 <p>1. Java is a?</p>
 <input type="radio" name="q1" value="a"> Programming Language<br>
 <input type="radio" name="q1" value="b"> Operating System<br>

 <p>2. HTML stands for?</p>
 <input type="radio" name="q2" value="a"> Hyper Text Markup Language<br>
 <input type="radio" name="q2" value="b"> High Text Machine Language<br>

 <p>3. CSS is used for?</p>
 <input type="radio" name="q3" value="a"> Styling<br>
 <input type="radio" name="q3" value="b"> Database<br><br>

 <button type="submit">Submit Quiz</button>

 </form>
</div>

</body>
</html>
```

QuizServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/QuizServlet")
public class QuizServlet extends HttpServlet {

 protected void doPost(HttpServletRequest request, HttpServletResponse response)
 throws ServletException, IOException {

 response.setContentType("text/html");
 PrintWriter out = response.getWriter();

 String name = request.getParameter("name");
 String q1 = request.getParameter("q1");
 String q2 = request.getParameter("q2");
 String q3 = request.getParameter("q3");

 int score = 0;

 if ("a".equals(q1)) score++;
 if ("a".equals(q2)) score++;
 if ("a".equals(q3)) score++;

 out.println("<html><body style='font-family:Arial;background:#f2f6ff;'>");
 out.println("<div style='width:400px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");
 out.println("<h2 style='color:#004aad;'>Quiz Result</h2>");
 out.println("<p><b>Name:</b> " + name + "</p>");
 out.println("<p><b>Your Score:</b> " + score + "/3</p>");
 out.println("</div>");
 out.println("</body></html>");
 }
}
```

Output :





RESULT :

Thus, the Online Quiz Application using Servlet is successfully developed. The application accepts user inputs through an HTML form, processes the quiz answers using a Servlet, and displays the result on the browser.